

Root Finding and Minimization

Optimization and Learning for Robot Control

RCI Lab

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Previously:

- Stability of equilibria
- Discrete-time simulation
- Forward Euler / Backward Euler
- Runge-Kutta

Today:

- Notation
- Root finding
- Newton's method
- Minimization

Let:

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

Linearization:

$$f(x + \delta x) \approx f(x) + J_f(x)\delta x$$

where

$$J_f(x) = \frac{\partial f}{\partial x}$$

Chain rule:

$$f(g(x)) \approx f(g(x)) + J_f(g(x))J_g(x)\delta x$$

Root Finding

Goal:

Find x^* such that $f(x^*) = 0$

Examples:

- Equilibrium of continuous system
- Backward Euler step

Closely related: fixed point

$$x^* = f(x^*)$$

Fixed-Point Iteration

Iterative scheme:

$$x_{k+1} = f(x_k)$$

Converges if:

$$\rho(J_f(x^*)) < 1$$

Only works if fixed point is stable.

Convergence can be slow.

Newton's Method

Linearize:

$$f(x + \delta x) \approx f(x) + J_f(x)\delta x$$

Set approximation to zero:

$$f(x) + J_f(x)\delta x = 0$$

Solve:

$$J_f(x)\delta x = -f(x)$$

Update:

$$x \leftarrow x + \delta x$$

Quadratic Convergence

If:

- f smooth
- $J_f(x^*)$ invertible
- Initial guess close enough

Then:

$$\|x_{k+1} - x^*\| \leq C \|x_k - x^*\|^2$$

Quadratic convergence.

Example: Backward Euler

Backward Euler:

$$x_{k+1} = x_k + hf(x_{k+1})$$

Define residual:

$$R(x_{k+1}) = x_{k+1} - x_k - hf(x_{k+1})$$

Solve:

$$R(x_{k+1}) = 0$$

Using Newton:

$$J_R \delta x = -R$$

Minimization

Problem:

$$\min_x f(x)$$

First-order necessary condition:

$$\nabla f(x^*) = 0$$

Now we have root finding:

$$\nabla f(x) = 0$$

Apply Newton.

Newton for Minimization

Second-order expansion:

$$f(x + \delta x) \approx f(x) + \nabla f(x)^T \delta x + \frac{1}{2} \delta x^T H(x) \delta x$$

where

$$H(x) = \nabla^2 f(x)$$

Set gradient to zero:

$$\nabla f(x) + H(x) \delta x = 0$$

$$H(x) \delta x = -\nabla f(x)$$

$$x \leftarrow x + \delta x$$

Sufficient Conditions

Necessary condition:

$$\nabla f(x^*) = 0$$

Sufficient for local minimum:

$$\nabla^2 f(x^*) \succ 0$$

(positive definite Hessian)

If Hessian positive definite everywhere:

f strongly convex

Global minimum.

Regularization (Damped Newton)

If Hessian not positive definite:
Modify:

$$H_{\lambda} = H + \lambda I$$

Solve:

$$(H + \lambda I)\delta x = -\nabla f(x)$$

Ensures descent direction.
Also called damped Newton.

Gradient Descent Comparison

Gradient descent:

$$x_{k+1} = x_k - \alpha \nabla f(x_k)$$

$$\alpha > 0$$

Descent if:

$$\nabla^2 f \succ 0$$

Slower than Newton but cheaper per iteration.

Takeaways

- Root finding and minimization are deeply connected.
- Newton has quadratic convergence.
- Requires solving linear system.
- Regularization ensures descent.